

Sujan Dora

AI Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2 years of experience building LLM-RAG based document QA and production-ready AI solutions. Delivered systems that improved document QA accuracy by ~20%, reduced MFA verification load by ~25%, and optimized ID authentication pipelines, with hands-on experience in Python, PyTorch, TensorFlow, OpenCV, and AWS. Recent graduate with strong foundation in LLMs and RAG systems, model evaluation, and end-to-end ML workflows.

EDUCATION

Missouri State University
M.S. Computer Science – Data Science

GPA: 3.63
Dec 2025

TECHNICAL SKILLS

Programming Languages	Python, SQL
ML Frameworks & Libraries	TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers
LLM & NLP Tools	LangChain, LangGraph, LangSmith, LlamaIndex, Agents, Multi-Agents, LoRA, QLoRA, OpenAI, Ollama, RAG Systems
ML Specializations	NLP, Computer Vision, Deep Learning, Anomaly Detection
MLOps & Deployment	MLflow, Git, GitHub Actions, CI/CD Pipelines, Model Deployment
Cloud & Infrastructure	AWS (Lambda, S3, EC2, ECR, API Gateway)
Databases & AI Infrastructure	PostgreSQL, MySQL, Milvus, Weaviate, FAISS, MongoDB
Web & DevOps	FastAPI, Docker Containerization

WORK EXPERIENCE

AI/ML Engineer
Grootan Technologies

Chennai, India
July 2023 December 2023

- Implemented a self-correcting RAG system using LangChain and Mistral 7B, improving answer reliability and mitigating context errors in document-level QA.
- Built a hybrid retrieval workflow combining dense embeddings and BM25 search, enhancing relevance and coverage of retrieved documents.
- Developed an LLM-as-Judge evaluation pipeline to assess faithfulness and relevance across hundreds of QA examples, ensuring production-ready reliability.
- Introduced batched query extraction, reducing LLM calls by ~80% per query and enabling faster, reliable production inference.
- Conducted applied ML research resulting in IEEE conference publication investigating bias detection in grading systems, demonstrating ability to pursue novel research questions, design experiments, and contribute to academic literaturefoundational skills for research in distributed ML over decentralized infrastructure.
- Experimented with multiple model architectures (YOLO, MiDaS, GANs) for counterfeit detection applying transfer learning and synthetic data generation, showcasing hands-on applied ML engineering with experimental frameworks for testing novel approaches in resource-constrained and heterogeneous environments.
- Designed an ML-based anomaly detection system for MFA workflows, reducing redundant verification prompts by 25%, improving user experience while maintaining security coverage.

AI/ML Intern
Grootan Technologies

Chennai, India
September 2022 July 2023

- Built a National ID authentication system for detecting counterfeit holograms using YOLO object detection, MiDaS depth estimation, and GAN-generated synthetic samples to improve robustness.
- Optimized the hologram verification pipeline by eliminating redundant processing, reducing per-card detection time from 7 minutes to 3 minutes.
- Applied transfer learning to fine-tune pre-trained models for limited datasets, achieving reliable counterfeit detection across varied hologram designs.
- Containerized and deployed solutions using Docker and AWS, ensuring scalable production-ready pipelines for anomaly detection, identity verification, and document QA systems.

Data Science Intern
iNeuron.ai

Bengaluru, India
November 2021 March 2022

- Developed an end-to-end credit card default prediction project using Docker, CircleCI, and Heroku, achieving 87% model accuracy on simulated customer transactions.
- Built a CI/CD pipeline to deploy predictive models to AWS production, enabling real-time predictions.
- Conducted exploratory data analysis on 30,000+ customer records to identify key features that improved model performance.
- Implemented data validation and preprocessing pipelines, reducing manual data preparation effort.

PERSONAL PROJECTS

Wine Quality Prediction - End-to-End MLOps Pipeline

- Built an end-to-end ML pipeline using ZenML and MLflow, deploying a REST API on AWS Lambda with Docker for automated training and inference.
- Implemented CI/CD with GitHub Actions to automate model retraining and deployment to AWS (S3, ECR, API Gateway), ensuring reliable and repeatable releases.

BioMedical Research Assistant Multi-Agent LLM System

- Fine-tuned Llama-3.1-8B using QLoRA on PubMedQA data with 4-bit quantization, gaining hands-on experience in parameter-efficient LLM training for biomedical QA.
- Built a multi-agent research workflow using CrewAI and PubMed retrieval via Model Context Protocol, with a Streamlit interface for interactive querying.

SafeSpace 2.0 AI Mental Health Therapist

- Developed an AI-powered mental health companion with empathetic conversational support, crisis detection, and location-aware therapist recommendations.
- Implemented a tool-using LLM agent (LangGraph / REAct) with multi-channel frontends (Streamlit web chat + Twilio WhatsApp) and integrated APIs for real-time actionable outputs.

PUBLICATIONS & RESEARCH

Dora, S. et al. (2021). "Detecting Bias in the Relative Grading System Using Machine Learning." *IEEE International Conference on Computing and Communications Technologies*.